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METHOD AND APPARATUS FOR MAKING DENTAL CROWNS

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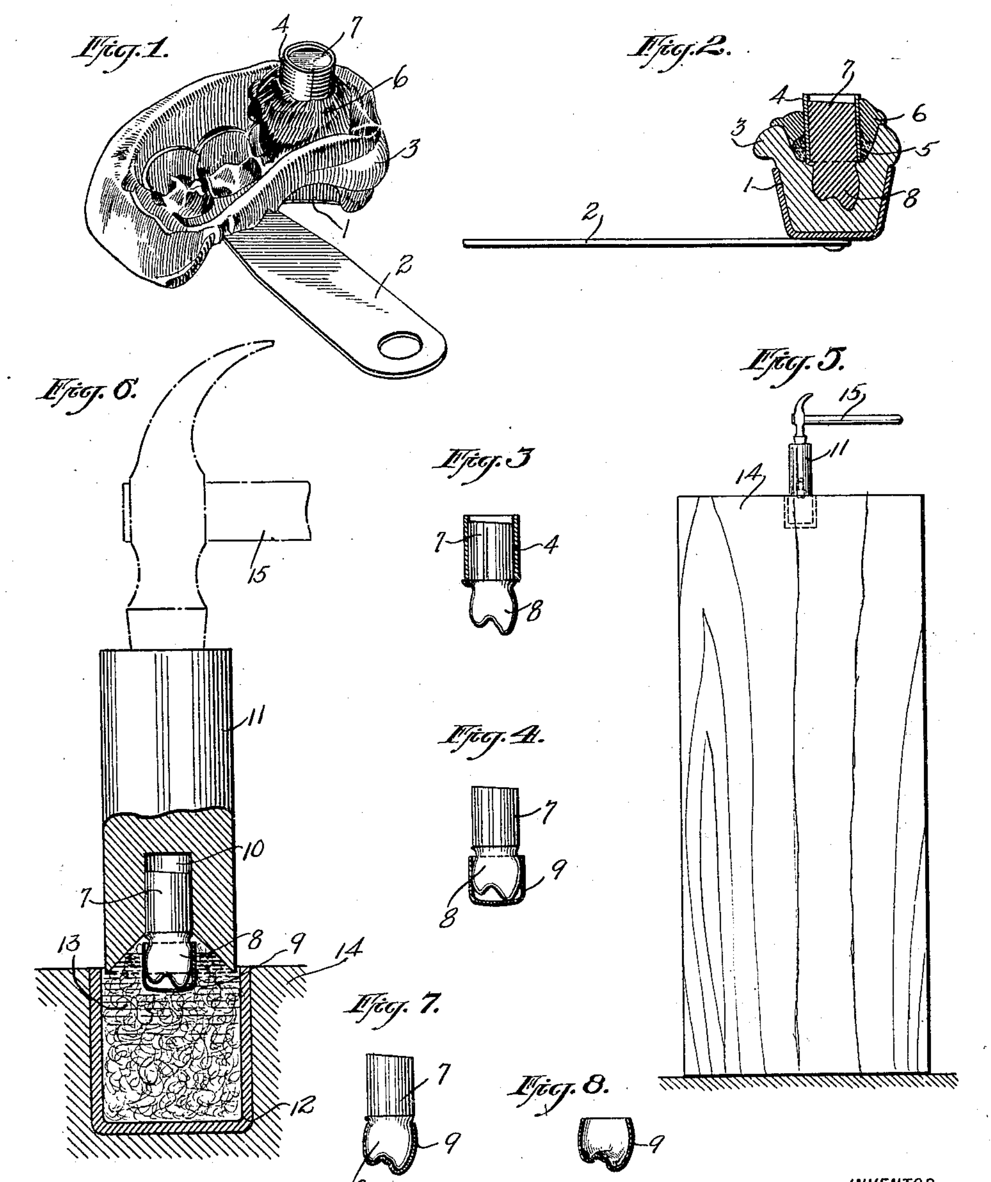

WITNESSES

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METHOD AND APPARATUS FOR MAKING DENTAL CROWNS.

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To all whom it may concern:

Be it known that I, JOSEPH M. FLANIGAN, a citizen of the United States, and a resident of Shamokin, in the county of Northumberland and State of Pennsylvania, have invented a new and Improved Method and Apparatus for Making Dental Crowns, of which the following is a full, clear, and exact description.

This invention relates to a method and apparatus for making dental crowns. An object of the invention resides in the provision of means whereby crowns for teeth of gold or other metal may be very quickly and easily constructed with a minimum number of operations, thereby saving a great deal of time and labor and rendering the operation more economical than has hitherto been the case.

Another object resides in the provision of means whereby a very true and exact crown can be made of a tooth of any configuration or articulation.

A further object resides in the provision of a method in which the steps are reduced to a minimum and in which the ease of manipulation of the various steps and processes is considerably enhanced.

A still further object resides in the particular sequence of operations and in the arrangement and construction of the parts hereinafter described and claimed and shown in the accompanying drawings.

The invention is shown in the drawings, of which

Figure 1 is a perspective view of an impression of a bite of the contour of certain teeth for which it may be desired to make a crown.

Figure 2 is a transverse section taken through the device shown in Fig. 1.

Figure 3 is a view of a metal tooth moulded and showing the matrix in position.

Figure 4 is a side view of a metal tooth with a gold crown placed thereover but not formed thereon.

Figure 5 shows the apparatus used to shape or form the gold crown in accordance with the contour of the tooth.

Figure 6 is an enlarged detail view of the apparatus shown in Fig. 5.

Figure 7 is a view showing the metal tooth with the crown shaped thereon.

Figure 8 is a section of the crown removed from the metal tooth.

The process and the apparatus set forth in the drawings and hereinafter described are preferred applications of my invention but various changes may be made in the manner of performing or carrying on the operation and in the construction and arrangement of the parts without departing from the general spirit of the invention as described and set forth herein.

In the first step taken to produce a gold crown for any given tooth, it is necessary for any person familiar with the operation of taking impressions of teeth to place in the patient's mouth a device which may be a mold receptacle, such as 1, having a handle 2. Within this receptacle 1 is placed a plastic material of any desired character in which is formed an outline of the contour of certain teeth when the patient sinks his teeth into the plastic material 3. Assuming that it is desired to produce a crown of gold or other material for one of the teeth of which impressions have thus been obtained, the first step is to place over the impression of that particular tooth a matrix 4, which as shown is in the form of a shell of any desired metal, but which can be made in the form of a sheet and then bent around to form a cylindrical shell. The ends of this shell or sheet may be lapped over each other or may be made of exactly the right size. Furthermore, instead of being made in a sheet the shell may be made in cylindrical form with the ends suitably connected or welded together into a cylinder of the proper diameter. The ends of this matrix or shell need not be connected or welded but may be placed with their edges abutting and then this joint may be sealed by the gold which is later placed around the matrix. This cylinder or shell or matrix is placed over the impression of the tooth and around it is placed a composition of metal, such as 5, and clay 6, to seal the shell in position. After this shell or matrix is firmly fixed in position, molten metal 7 of any suitable composition is poured into the matrix and fills the impression of the tooth as well as the space within the shell. There is, therefore, formed in the impression space a model or duplicate 8 of the tooth for which it is desired to produce a crown. The matrix, with the metal tooth, is then removed from the impression or mold and is in the form shown in Fig. 3. The matrix is then removed from the metal tooth and a gold crown, such as 9 in Fig. 4, is placed over the portion 8 of this metal tooth and is shaped thereon in accordance with the contour of the tooth. The portion 7 of the tooth, which I may call the stem portion, is then placed in a suitable bore such as 10 in a swaging block 11. The stem is so placed within the bore that the tooth portion 8 and the gold crown 9 extend into a receptacle 12. This receptacle 12 is of any suitable material, preferably metal, and contains a mass of some resilient cushioning material 13, which is preferably wet cotton. This receptacle 12 for convenience may be disposed in a suitable recess in the top of a large block or support 14 which may act in the nature of an anvil. The size of the opening of the receptacle 12 is substantially the same as the outer diameter of the swaging block 11, and, therefore, to properly shape the gold crown on to the tooth 8 it is merely necessary to tap or strike the swaging block 11 with any suitable instrument, such as a hammer 15, a few times, forcing the swaging block with the tooth and the crown into the receptacle 12. The pressure of the cotton against the crown will force it and form it against the tooth 8 in accordance with the configuration of this tooth. The flexibility and resiliency of the cotton will permit this intimate formation of the crown to be effective, while at the same time, this flexibility will prevent any undue pressure from injuring either the tooth or the crown, and in this way the cotton acts not only as a means for applying pressure to the corwn but as a means of cushioning the blows or impacts to which the crown and tooth are subjected. It is found that with but several taps of the hammer the gold crown, which is made of soft flexible material, is shaped on to the tooth. The tooth and the crown are then removed from the swaging block 11 and the crown can, by reason of its flexibility, be readily withdrawn from the tooth. Before taking the gold crown off the metal tooth 8, this tooth acts as means for firmly and conveniently holding the crown without distorting it during the time it is being polished. This metal tooth, which is a sort of a model of the patient's tooth, may be preserved, if desired, for record or for future use, or, being made of fusible metal, can be readily melted for remoulding again.

It will, therefore, be seen that I have provided a simple, efficient apparatus and method whereby crowns for teeth can be very quickly and easily made with a minimum expenditure of time and labor and in a very economical manner. The gold cap or crown can also be used to place over the tooth in which it is desired to hold or retain medicine in the treatment of a nerve or sensitive part. This treatment is particularly sanitary since the tooth can be covered to hold the medicine in place. The gold caps or crowns can, of course, be soldered permanently in place after the final treatment.

What I claim is:

1. A dental mold which comprises a plastic impression of a tooth to be crowned, a hollow shell-like matrix disposed over the impression of the tooth, and sealing material disposed around said matrix to seal the matrix in position over the impression whereby liquid metal can be poured into the matrix and the impression to form a mold for a metal tooth and a stem therefor.

2. A matrix for use in forming metal teeth, which comprises a sheet of flexible metal capable of being folded up in the form of a cylinder to be placed over the recess or impression of a tooth in a bite of plastic material.

3. A method of forming a metal tooth which comprises forming an impression of the desired tooth in a plastic bite, placing a cylindrical matrix over this impression to seal it off from the rest of the bite, and then pouring moulten metal into the matrix and the impression to form a metal tooth having a stem and a tooth portion.

4. The method of forming a metal tooth, which comprises forming an impression of the desired tooth in a plastic bite, placing a cylindrical matrix over the impression to seal it from the remainder of the bite, placing sealing material around the matrix to hold it in position and to seal the joint between the matrix and the impression of the tooth, and then pouring liquid metal in through the matrix into the impression to form a tooth and a stem portion thereon.

JOSEPH MICHAEL FLANIGAN.